

B.N.M. Institute of Technology

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Vidyayāmruthamashnuthe

INNOVATIVE PROJECT LAB

(23ECE139)

(Project Based Learning Course)

PROJECT REPORT

ON

**“AFFORDABLE AND PORTABLE INCUBATOR FOR PREMATURE
BABIES”**

by

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2024-2025

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CERTIFICATE

Certified that the IPL Project titled “Affordable and Portable Incubator for Premature Babies” carried out by **Shreya Mouli G N(1BG23EC100)**, **Poorvika M R(1BG23EC078)**, and **Sangeeta S Gatti(1BG23EC094)**, bonafide students of **III semester**, during the year **2024-2025**, for the fulfillment of the academic requirements for the Project Based Learning Course Innovative Project Lab (**23ECE139**).

The Project report has been approved as it satisfies the academic requirements in respect to the course.

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Place-BNM Institute of Technology

Date-13/12/2024

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ABSTRACT

The leading cause of child mortality globally is prematurity. These deaths mostly occur due to unavailable or high cost of keeping babies in clinical incubators. The aim of this project is to develop a cheap and portable wireless incubator for premature babies. The project focuses on reducing the causes of preterm death and, as such, incorporates measures to minimize such factors. Hence, the incubator's other functions are to reduce Sudden Infant Death Syndrome (SIDS) and malaria. It has sensors to monitor vital environmental conditions of the incubator. It sends this information to the paediatrician to keep track of conditions that might affect the health of the baby. In order to maintain the set temperature continuously, the incubator can be powered from the grid and a battery used as an alternative. The incubator can maintain a constant temperature. A basket is used as the incubator's bassinette. The electronic components are cased and attached to the outside of the basket which makes the incubator portable. The baby will be kept in the basket with the required dimension to reduce SIDS by avoiding instances.

4. INTRODUCTION

4.1 The Problem Statement

The World Health Organization in 2015 estimated that 1 million among 15 million of premature babies do not survive. Globally, the leading cause of child mortality is prematurity [1]. Half of babies born two months early or less die in low-income settings due to lack of feasible, low-cost care such as warmth, breastfeeding support, and basic care for infections [1]. In India, neonatal mortality accounts for nearly 50% of infant mortality [2]. Also, about 18% of all Sudden Infant Death Syndrome (SIDS) deaths are that of premature infants [3]. Additionally, the highest prevalence of malaria cases, particularly deaths, exist among children below the age of 5 (90%), representing the main cause of child mortality in Africa [4].

4.2 Motivation

Mothers from low-income settings cannot afford the cost of keeping their premature babies in clinical incubators. This proposed solution will be cost-effective for such mothers. The project is also aimed to reduce SIDS caused by babies rolling to the side and sleeping on their stomachs. It will further be designed to reduce baby mortality caused by malaria.

4.3 Proposed Solution

In order to address some of the factors leading to infant mortality, help save lives by providing warmth to premature babies and improve on the existing premature incubators, this project is aimed to design and implement a simple, cheap and portable premature baby incubator that can maintain the required temperature of 37°C continuously.

4.4 Contribution

This project seeks to bring the following contributions to the problems of premature infant death:

- Provide continuous warmth for premature babies at an affordable cost in a portable and in a family-friendly setting.
- The incubator will help reduce preterm death caused by malaria.
- The power consumption from the battery will be monitored and the battery can either be recharged through solar power or electricity from the grid.
- Sensors will be incorporated to monitor environmental conditions, and the information will be visible to both mother through a display and wirelessly to the paediatrician.

5. LITERATURE REVIEW

5.1 Background to the research

Causes of death for premature babies can be as a result of many factors, and one is the inability to control and maintain their body temperature [1]. Every year in developing countries, approximately 20 million premature and low birth weight babies are born, 4 million die within their first month and such deaths occur because traditional incubators are unreliable or inaccessible [1]. Mothers from low-income settings cannot afford the cost of keeping their premature babies in clinical incubators.

SIDS is typically associated with a sleep cycle, and risk factors such as prone/face-down sleep, bed sharing, soft bedding, and over-bundling in the sleep area. As a result, a tradition that dates back to the 1930's has been established to give all children, regardless of background, an equal start to life through the provision of a cardboard box to expectant mothers with a mattress in the bottom that becomes a baby's first bed. Through this, Finland has one of the lowest infant mortality rates [2]. Malaria is another cause of infant mortality that needs critical attention when it comes to infant mortality. According to World Health Organization [6], "Every 2 minutes, a childish of malaria. And each year, more than 200 million new cases of the disease are reported. Although countries have dramatically reduced the total number of malaria cases and deaths since 2000, progress in recent years has stalled. Worryingly, in some countries, malaria is on the rise". As a result, infants most especially premature babies in this instance should not be exposed to mosquito bites. This can be achieved by the usage of insecticide-treated mosquito net.

5.2 Survey on related works

A major contributor to neonatal mortality and morbidity throughout the world is hypothermia which is clinically defined as body temperature below 36.5 degrees Celsius [4]. For neonatal survival, a well-regulated thermal environment is essential [5]. The purpose of temperature monitoring is to keep the child in a thermo neutral environment [6]. Even though advanced incubators currently exist, they are costly to meet the needs of developing countries [4].

5.2.1 Embrace Infant Warmer

Embrace Innovation is a medical innovation company with the goal of creating and supplying emerging markets with advanced, affordable, and reliable medical devices. Their first product is a low-tech infant warmer; a portable, low-cost incubator to reduce the risk of death in preterm infants in developing countries. The Embrace Infant Warmer is small and light and appears like a tiny sleeping bag and is easily transported to rural villages

5.2.2 Inflatable Incubator

A graduate from Loughborough University, James Roberts created a manually inflatable incubator which is warmed by using ceramic heating elements. It has a screen that displays the temperature and humidity dependent on the baby's age, an alarm that sounds if the required temperature changes and a phototherapy unit for babies suffering from jaundice. It is similar to a modern incubation system and costs £250 to manufacture and test. The device can be collapsed for transportation and runs off a battery that lasts 24 hours, in case of power unreliability

5.2.3 Smart Infant Incubator

The development of a smart infant incubator monitoring and control system that uses IoT technology to customize the care and monitoring of premature and critically ill newborns. As the system is interfaced with advanced sensors, actuators, and connective solutions to create a smart and responsive incubator environment. The system adopts with some key components such as temperature and humidity sensor for precise check of environmental conditions, gas sensor, heartbeat sensor, sound sensor and accelerometer to monitor the infant's conditions inside the incubator and actuators such as Peltier module and humidifier are integrated to adjust the environmental conditions according to the threshold range on the processed sensor data

5.2.3 Mothers' involvement in caring for their premature infants

Advances in technology have resulted in increasing survival rates even for extremely premature infants. While sophisticated medical management is vital to infant survival, research has found that social factors and care giving processes are important predictors of infants' later outcome. Consequently, evidence is accumulating to demonstrate the fundamental role of mothers and families to the optimal developmental outcome of premature infants

5.2.4 The incubation of a social movement

This article explores health-based activism on the part of the US ‘parents of preemies’ movement, a mutual-help network mobilized around babies born precariously early and acutely dependent upon life-support incubators. The movement articulates two meta-agendas for parental empowerment: (1) the quest to access/exercise greater participatory inclusivity vis-a`-vis the preterm baby within the biomedical domain; and, (2) the quest to secure/command greater representational authority over the preterm baby within the public domain. Seen in terms of the erosion of the status quo, it can be argued that the movement’s tangible and intangible aims to chip away at these traditions have been softly revolutionary: heralding new working partnerships between medical practitioners and patients’ families; radical shifts in the technological consciousness and competences of premie parents

6. METHODOLOGY

6.1 Standard incubator specifications

This project was designed by taking into consideration the standard incubator specifications. The standard incubator has over temperature protection and as such has an automatic cut off of heater at 39.3 °C. It has audio and visual alarms. These alarms operate when the baby's temperature is Low/High ($\pm 5\%$), air temperature is Low/High ($\pm 5\%$), temperature probe failure, system failure, heater failure, low battery and main power failure. It also displays the baby's temperature, air temperature, set temperature and AC power on a Liquid Crystal Display (LCD).

6.2 COMPONENTS REQUIRED

1.ESP32



Fig 6.1 ESP32

The ESP32 is a versatile and powerful micro controller developed by Espressif Systems, widely used in IOT and embedded systems applications. Equipped with dual-core processors, built-in Wi-Fi, and Bluetooth capabilities, it offers robust connectivity options for various smart devices. It features multiple GPIO pins, analog inputs, and digital communication interfaces like UART, I2C, and SPI, enabling seamless integration with sensors, actuators, and other peripherals. The ESP32 also supports advanced functionalities like low-power modes, touch sensing, and audio processing, making it suitable for diverse applications ranging from smart home automation to industrial control systems. Its affordability, open-source ecosystem, and extensive community support further enhance its popularity among developers and hobbyists.

2. DHT11

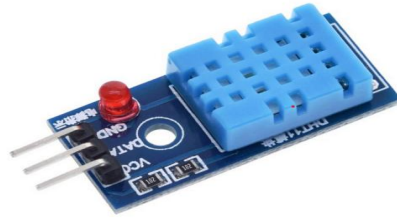


Fig 6.2 DHT11

The DHT11 is a low-cost digital sensor used for measuring temperature and humidity, making it ideal for various IoT and environmental monitoring applications. It operates on a single wire and provides calibrated digital output, ensuring ease of use and integration with micro-controllers like Arduino, Raspberry Pi, or ESP32. The DHT11 has a temperature measurement range of 0–50°C with $\pm 2^\circ\text{C}$ accuracy and a humidity range of 20–90% RH with $\pm 5\%$ accuracy. Its low power consumption, compact size, and per-calibrated nature make it suitable for projects requiring basic environmental sensing, such as home automation, weather stations, and agricultural monitoring systems. However, its moderate precision and slower response time may limit its usage in applications demanding high accuracy or rapid data updates.

3. MQ2 GAS SENSOR

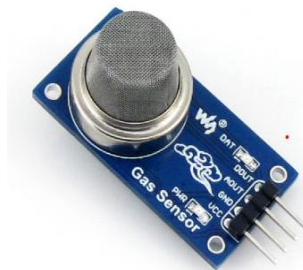


Fig 6.3 MQ2 Gas Sensor

The MQ2 gas sensor can be an essential component in infant incubators for premature babies, enhancing safety by detecting the presence of harmful gases such as smoke, LPG, methane, and carbon monoxide. It operates by sensing changes in gas concentrations and providing an analog or digital output for real-time monitoring. Integrating the MQ2 sensor into an infant incubator helps ensure a safe environment by triggering alarms or ventilation systems if

hazardous gas levels are detected. This is particularly critical for premature babies, who are more vulnerable to respiratory issues. The sensor's low cost, sensitivity, and compatibility with micro-controllers like ESP32 or Arduino make it a practical choice for inclusion in advanced healthcare monitoring systems.

4.LCD Display



Fig 6.4 LCD Display

An LCD (Liquid Crystal Display) is a widely used electronic display technology known for its low power consumption, clear visuals, and lightweight design. It works by manipulating liquid crystals to control the passage of light, creating images or text on the screen. LCD are available in various sizes and types, from simple character-based modules (like 16x2 or 20x4 displays) to high-resolution graphical displays. They are commonly interfaced with micro-controllers like Arduino, Raspberry Pi, or ESP32 for displaying real-time data, such as sensor readings, system status, or user interfaces. Their versatility and cost-effectiveness make them suitable for applications ranging from consumer electronics to industrial systems and IOT devices.

5.Relay



Fig 6.5 Relay

A relay is an microelectronic or solid-state device used to control high-power or high-voltage circuits with a low-power signal. It acts as a switch, allowing electrical isolation between the

controlling circuit and the load circuit. Relays are commonly used in applications where direct control of high-current devices, such as motors, lights, or heaters, is required. In IOT and embedded systems, they are often interfaced with micro-controllers like Arduino or ESP32 to enable automation and remote control. Their versatility, reliability, and ability to handle both AC and DC loads make them indispensable components in home automation, industrial control, and safety systems.

6. Buzzer



Fig 6.6 Buzzer

A buzzer in an infant incubator for premature babies serves as a critical alert mechanism, enhancing safety and ensuring prompt response to potential issues. It is used to provide auditory alarms for various conditions, such as abnormal temperature or humidity levels, equipment malfunctions, or critical health parameters detected by integrated sensors. This immediate feedback allows caregivers to take necessary actions quickly, ensuring the infant's environment remains stable and conducive to recovery. The buzzer's simplicity, reliability, and compatibility with micro-controllers like ESP32 make it a valuable component in improving the overall functionality and safety of the incubator system.

7. Thermostat



Fig 6.7 Thermostat

A thermostat is a device used to regulate temperature by switching heating or cooling systems on or off based on preset thresholds. In many electronic thermostats, terminals like K0 and K1 are used as control outputs for external devices such as relays or actuators. These terminals enable the thermostat to communicate and manage connected systems effectively.

The VCC terminal provides power to the thermostat's internal circuitry, typically operating on low voltage (e.g., 3.3V or 5V). The GND terminal serves as the common ground reference for the power supply and signal connections. These connections ensure reliable operation, allowing the thermostat to accurately monitor temperature, execute control actions, and integrate seamlessly into larger systems like HVAC or smart home networks.

6.2.2 Block Diagram

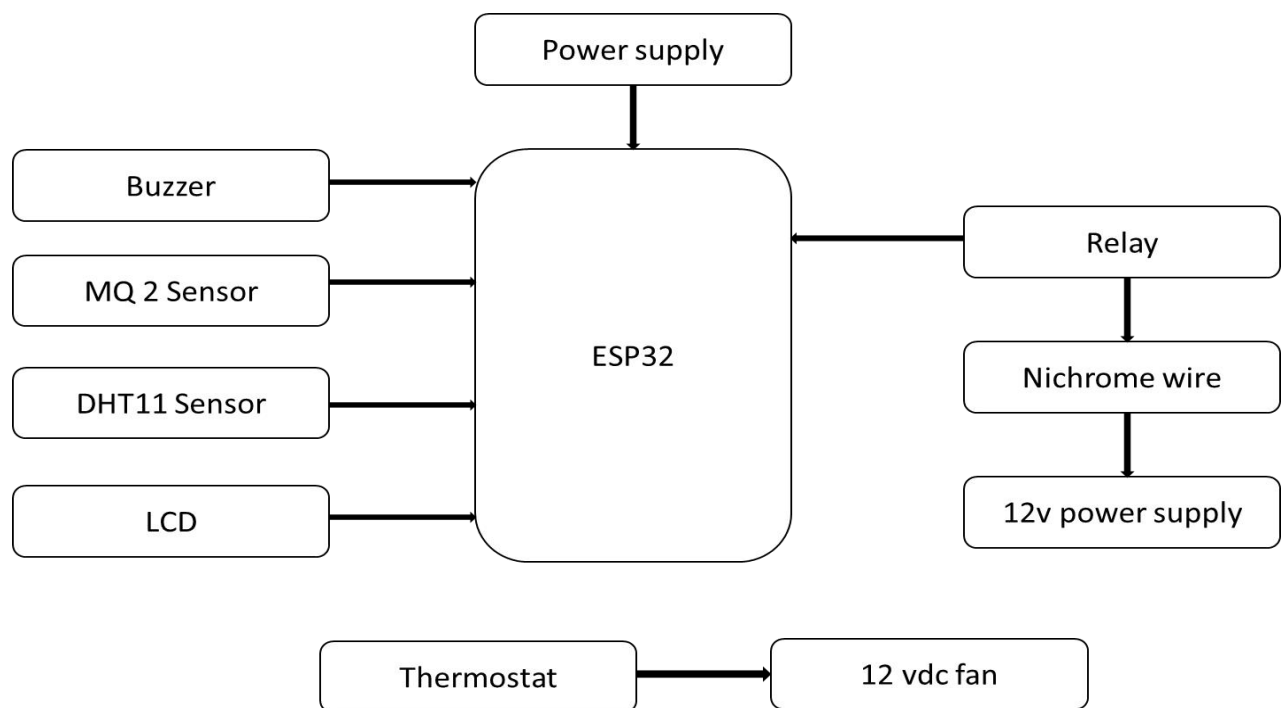


Figure 6.2: Block diagram of incubator system.

6.2.3 Cost of the Project

Components	Price in Rupees
ESP 32	300/-
Battery(9V)	25/-
Relay board(5V)	50/-
DHT11sensor	75/-
Metal Oxide gas sensor	90/-
SVDC fan	60/-
LCD	100/-
Battery level Indicator	80/-
Nichrome wire	25/-
Buzzer	15/-
Total	820/-

The cost of the project estimated above is an approximate cost and can vary slightly.

7. CONCLUSION

This premature baby incubator is cheap, wireless, portable and maintains the needed temperature continuously. It can be powered with power supply from the grid or battery. It comes with a battery level indicator to ensure provision of continuous energy to regulate the baby's temperature. It is designed to be family-friendly. It is a smart system that has sensors to detect temperature, humidity, toxic gas(smoke), and light intensity. It further displays the information to the mother via an LCD screen and wirelessly to the Pediatrician. The addition of insecticide treated mosquito net will help reduce baby mortality caused by malaria.

8. FUTURE DEVELOPMENTS

Several applications can be incorporated into the system to further improve it. One of the improvements is including weight sensing which is vital in detecting the growth of the baby. A 5kg load cell can be used in conjunction with HX711 to determine the weight of the baby which can be displayed on the LCD screen and also to the database for the Pediatrician. The load cell can be cased with wooden platform and the weight of the bassinette without the baby in, tiered. When the baby is put in the bassinette and placed on the weighing design, the weight of the baby will read on the LCD screen. The value displayed on the LCD screen is for the user(mother) and then transferred wirelessly to the Pediatrician. Noise detection can also be factored in to detect the cry of the baby and sound an alarm wirelessly to the mother. Breathing detection is another factor to improve this project. The breathing detector can be designed and placed under the mattress of the incubator. It will detect movement and when there is no movement detected, there will be an alarm to alert the mother.

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